A-A-50213A
18 May 1988
SUPERSEDING
A-A-50213
July 28, 1983

COMMERCIAL ITEM DESCRIPTION

CURVES, DRAFTING, REGULAR AND IRREGULAR:

CURVE SETS, DRAFTING, REGULAR AND IRREGULAR

The General Services Administration has authorized the use of this commercial item description if preference to Federal Specification GG-C-871.

This description covers drafting curves and drafting curve sets for use in general drafting and drawing work.

Salient characteristics:

Design. The individual item requirements shall be as specified herein. The thickness of the curves shall be 0.060 plus or minus 0.012 inch. All edges shall be square and free from nicks and burrs. The curve surface shall not vary from a true plane by more than 0.010 inch.

Plastic material. Methacrylic, extruded or cast acrylic.

Types. Types of curves shall be as shown below:

A = Type I - Curves, French (irregular), set or individual (see figure 1).
B = Type II - Curves, mechanical engineers' (irregular), set (see figure 2).
Type III - Curves, railroad (regular), set or individual.
C = Style 1 - With tangent (see tables I and II).
D = Style 2 - Without tangent (see table III).
E = Type IV - Curves, ship (irregular), set or individual (see figure 3).

Part number. The document part number for the curves, drafting, regular and irregular shall be as follows:

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A50213 - X - Y

- Y — Curve: Type I, II, and IV; 1 to 150 (see figures 1, 2, and 3)
  Type III (style 1); (see table I and II degrees or inches of radius
  Type III, (style 2); (see table III) inches of radius - 1 to 120
- X — Type curve: A = Type I
  B = Type II
  C = Type IV (style 1)
  D = Type III (style 2)
  E = Type IV
- A50213 — CID identification number (Do not include the revision letter or activity symbol.)

Patterns. Curves shall be furnished by pattern numbers (see applicable figure), in sets or individually. The size of the curves shall be the size regularly produced by manufacturers of the curves.

Marking. Each curve shall have the applicable pattern number (see applicable figure) stamped or otherwise permanently marked upon the item.

Case. A case for carrying and protecting the curves shall be furnished. The case shall be equipped with metal hinges and fasteners. When an individual curve is provided, it shall be in a clear plastic envelope.

TABLE I. Style 1: Railroad curves with tangent. Individual and sets of 41 and 55 curvatures based on degrees.

| (One inch = 100') | |
|---|---|
| 0° 15' = 229.18 inches | *7° 15' = 7.91 inches |
| *0° 30' = 114.59 inches | *7° 30' = 7.64 inches |
| 0° 45' = 76.39 inches | *7° 45' = 7.40 inches |
| *1° 00' = 57.30 inches | *8° 00' = 7.17 inches |
| *1° 15' = 45.84 inches | *8° 15' = 6.95 inches |
| *1° 30' = 38.20 inches | *8° 30' = 6.75 inches |
| *1° 45' = 32.74 inches | *8° 45' = 6.55 inches |
| *2° 00' = 28.65 inches | *9° 00' = 6.37 inches |
| *2° 15' = 25.47 inches | *9° 15' = 6.20 inches |
| *2° 30' = 22.92 inches | *9° 30' = 6.04 inches |
| *2° 45' = 20.84 inches | *9° 45' = 5.88 inches |
| *3° 00' = 19.10 inches | *10° 00' = 5.74 inches |



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TABLE I. Style 1: Railroad curves with tangent. Individual and sets of 41 and 55 curvatures based on degrees. (Continued)

| (One inch = 100') | | | |
|---|---|---|---|
| *3° 15' = | 17.63 inches | *10° 30' = | 5.46 inches |
| *3° 30' = | 16.37 inches | *11° 00' = | 5.22 inches |
| *3° 45' = | 15.28 inches | *11° 30' = | 4.99 inches |
| *4° 00' = | 14.33 inches | 12° 00' = | 4.78 inches |
| *4° 15' = | 13.48 inches | 12° 30' = | 4.59 inches |
| *4° 30' = | 12.73 inches | 13° 00' = | 4.42 inches |
| *4° 45' = | 12.07 inches | 13° 30' = | 4.25 inches |
| *5° 00' = | 11.46 inches | 14° 00' = | 4.10 inches |
| *5° 15' = | 10.92 inches | 14° 30' = | 3.96 inches |
| *5° 30' = | 10.42 inches | 15° 00' = | 3.83 inches |
| *5° 45' = | 9.97 inches | 16° 00' = | 3.59 inches |
| *6° 00' = | 9.55 inches | 17° 00' = | 3.38 inches |
| *6° 15' = | 9.17 inches | 18° 00' = | 3.20 inches |
| *6° 30' = | 8.82 inches | 19° 00' = | 3.03 inches |
| *6° 45' = | 8.49 inches | 20° 00' = | 2.88 inches |
| *7° 00' = | 8.19 inches | | |

* Set of 41 curves.

TABLE II. Style 1: Railroad curves with tangent. Individual and set of 55 curvatures based on inches of radius.

| | | | | |
|---|---|---|---|---|
| 3 | 12 | 23 | 38 | 100 |
| 3-1/2 | 13 | 24 | 40 | 110 |
| 4 | 14 | 25 | 45 | 120 |
| 4-1/2 | 15 | 26 | 50 | 130 |
| 5 | 16 | 27 | 55 | 140 |
| 6 | 17 | 28 | 60 | 150 |
| 7 | 18 | 30 | 65 | 160 |
| 8 | 19 | 32 | 70 | 170 |
| 9 | 20 | 34 | 75 | 180 |
| 10 | 21 | 35 | 80 | 190 |
| 11 | 22 | 36 | 90 | 200 |



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TABLE III. Style 2: Railroad curves without tangent. Individual and sets of 17, 30, and 50 curvatures based on inches of radius.

| Curve radius in inches | Set of 50 | Set of 30 | Set of 17 | Curve radius in inches | Set of 50 | Set of 30 | Set of 17 |
|---|---|---|---|---|---|---|---|
| 1-1/2 | X | X | - | 26 | X | X | - |
| 2 | X | X | - | 27 | - | - | X |
| 2-1/2 | X | X | - | 28 | X | X | - |
| 3 | X | X | - | 30 | X | X | X |
| 3-1/2 | X | X | - | 32 | X | - | - |
| 4 | X | X | - | 33 | - | - | X |
| 4-1/2 | X | X | - | 34 | X | - | - |
| 5 | X | X | - | 35 | - | X | - |
| 5-1/2 | X | X | - | 36 | X | - | X |
| 6 | X | X | - | 38 | X | - | - |
| 6-1/2 | X | - | - | 39 | - | - | X |
| 7 | X | X | - | 40 | X | X | - |
| 7-1/2 | X | - | - | 42 | - | - | X |
| 8 | X | X | - | 45 | X | X | X |
| 8-1/2 | X | - | - | 48 | - | - | X |
| 9 | X | X | - | 50 | X | X | - |
| 9-1/2 | X | - | - | 51 | - | - | X |
| 10 | X | X | - | 54 | - | - | X |
| 10-1/2 | X | - | - | 55 | X | - | - |
| 11 | X | X | - | 57 | - | - | X |
| 11-1/2 | X | - | - | 60 | X | X | X |
| 12 | X | X | X | 65 | X | - | - |
| 13 | - | - | - | 70 | X | - | - |
| 14 | X | X | - | 75 | X | - | - |
| 15 | - | - | X | 80 | X | - | - |
| 16 | X | X | - | 85 | X | - | - |
| 17 | - | - | - | 90 | X | - | - |
| 18 | X | X | X | 95 | X | - | - |
| 19 | - | - | - | 100 | X | - | - |
| 20 | X | X | - | 110 | X | - | - |
| 21 | - | - | X | 120 | X | - | - |
| 22 | X | X | - | | | | |
| 23 | - | - | - | | | | |
| 24 | X | X | X | | | | |
| 25 | - | - | - | | | | |



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Regulatory requirements. The offeror/contractor is encouraged to use recovered materials in accordance with Public Law 94-580 to the maximum extent practical.

Packaging.

Military packaging. Military packaging shall be specified for shipment to overseas Department of Defense (DoD) consignees and shall consist of the following:

Each carrying case with the curve set properly positioned therein and, as applicable, each individual curve contained within the plastic envelope, shall be preserved and packed to protect against physical damage and deterioration in a logistic environment characterized by the following:

1. Multiple handling during transportation and in-transit storage.
2. Shock, vibration and static loading forces during worldwide shipment by all transport means.
3. Uncontrolled warehouse storage for extended periods.
4. Weather exposure during shipment and in-transit transfers.
5. Stacking loads during shipment and storage.

Commercial packaging. Commercial packaging may be specified for shipments to DoD consignees within the Continental United States and to non-DoD consignees worldwide and shall consist of the following:

Each carrying case with the curve set properly positioned therein and, as applicable, each individual curve contained within the plastic envelope, shall be preserved and packed in accordance with the supplier's normal method for retail distribution. The method of protection and the shipping containers shall comply with the requirements of the Uniform Freight Classification and the National Motor Freight Classification.

Note: When a supplier's normal commercial packaging satisfies the requirements for military packaging, it may be used for that purpose.

Unitization. Individually packaged quantities of packed curves destined to a single addressee may be unitized into one or more unit loads.

Marking. In addition to any special or other identification marking required by the contractor or purchase order, each unit container, each intermediate container (when provided) and each exterior (shipping) container shall be marked with the item nomenclature, stock or part number, quantity per pack and contract number. Additionally, each exterior container shall be marked with the consignee address. Unitized loads of shipping containers all having equal quantities of the same item shall require no additional markings provided the shipping container markings can be read from two sides of the load. Markings shall be applied in a legible manner, in a contrasting color, of an appropriate size and of waterproof materials.



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Quality assurance. The curves shall be examined for the following defects. Presence of one or more defects shall be cause for rejection.

1. Design or dimensions not as specified.
2. Material not as specified.
3. Types of curves not as specified.
4. Patterns not as specified.
5. Packaging not as specified.
6. Marking of curves not as specified.
7. Case not as specified.

Certification of commercial item. The contractor shall certify that the product offered to the Government meets the salient characteristics of this description and that the product conforms to the producer's own drawings, specifications, standards, and quality assurance practices and is the same product as the manufacturer's current product offered by the contractor in the commercial marketplace for at least one year preceding the solicitation. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Small business. A small business that cannot meet the certification-of-commercial-item requirement of this document, may make an offer under this solicitation; and, if such small business is the apparent successful offeror, the Government may require that small business to submit a sample curve or set of curves. Any such curve or set of curves may be evaluated for this solicitation if time to contract award permits. Such evaluation would be to determine conformance to all requirements of this commercial item description and to determine equivalency to curves or set of curves available in the commercial marketplace in the areas of durability, human factors engineering, and safety. It should be noted that samples are not required unless specifically requested by the contracting officer. Note. The Government may elect not to evaluate a sample from an apparent low bidder for award for this solicitation if a determination of urgency is made by the contracting officer. The sample would then be evaluated prior to future solicitation for comparable items.

Note:

1. When this commercial item description is used for procurement, the commercial item certification clause must appear in the solicitation.



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2. Purchaser should specify type and style curves required, and whether in set or individual.
3. Figures. The figures show the shapes and pattern numbers for drafting curves which have been found acceptable; however, the figures are for illustration only and are not intended to preclude the furnishing of other designs which conform to the salient characteristics.

MILITARY INTERESTS:

Custodians

Army - ME
Navy - YD
Air Force - 99

Review Activities

Air Force - 82
DLA - GS

User Activities

Navy - MC

CIVILIAN AGENCY COORDINATING ACTIVITIES:

GSA - FOGC
VA - OSS

PREPARING ACTIVITY:

Army - ME

Project 6675-0814



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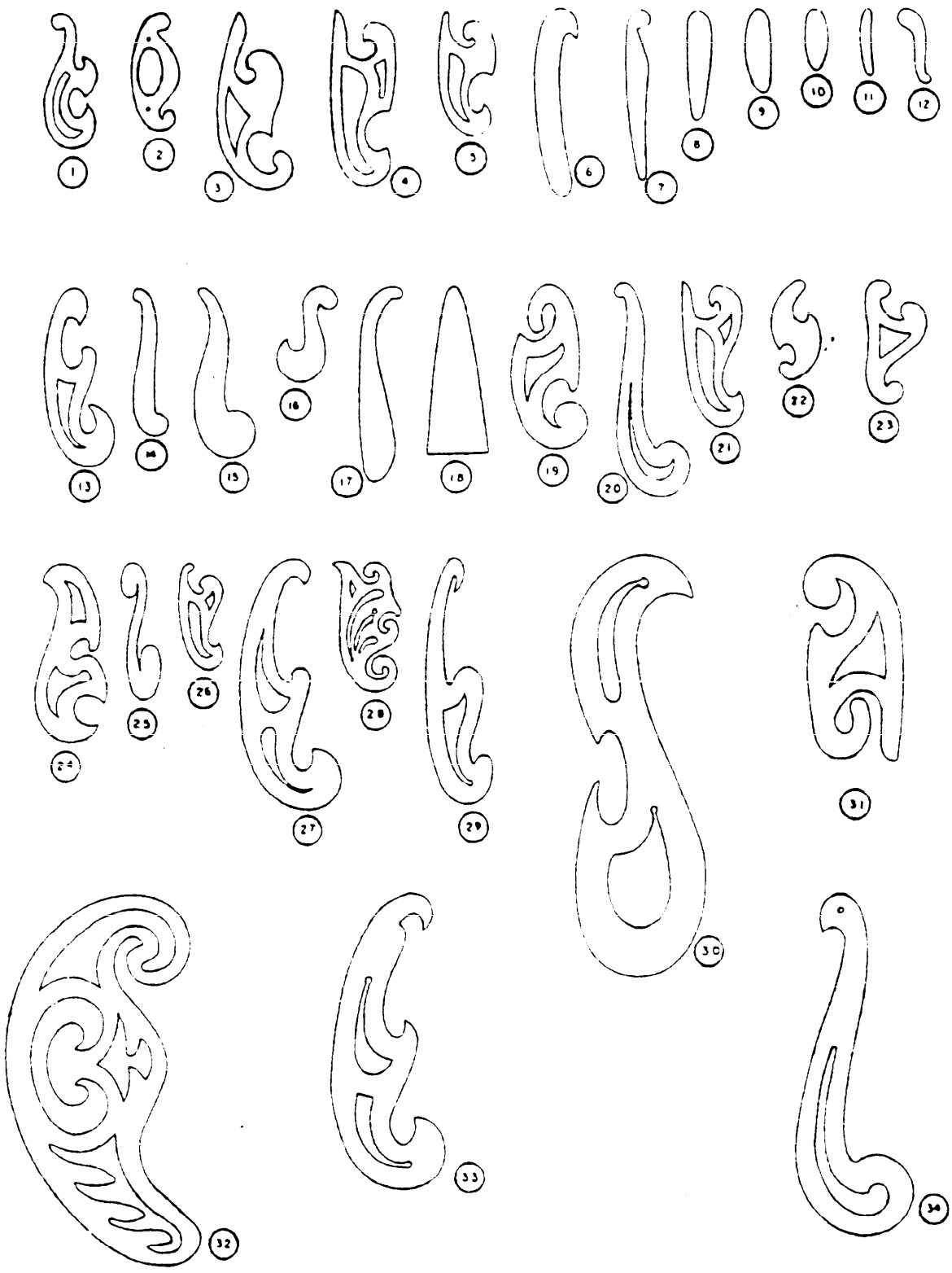


FIGURE 1. French (Irregular) curves.

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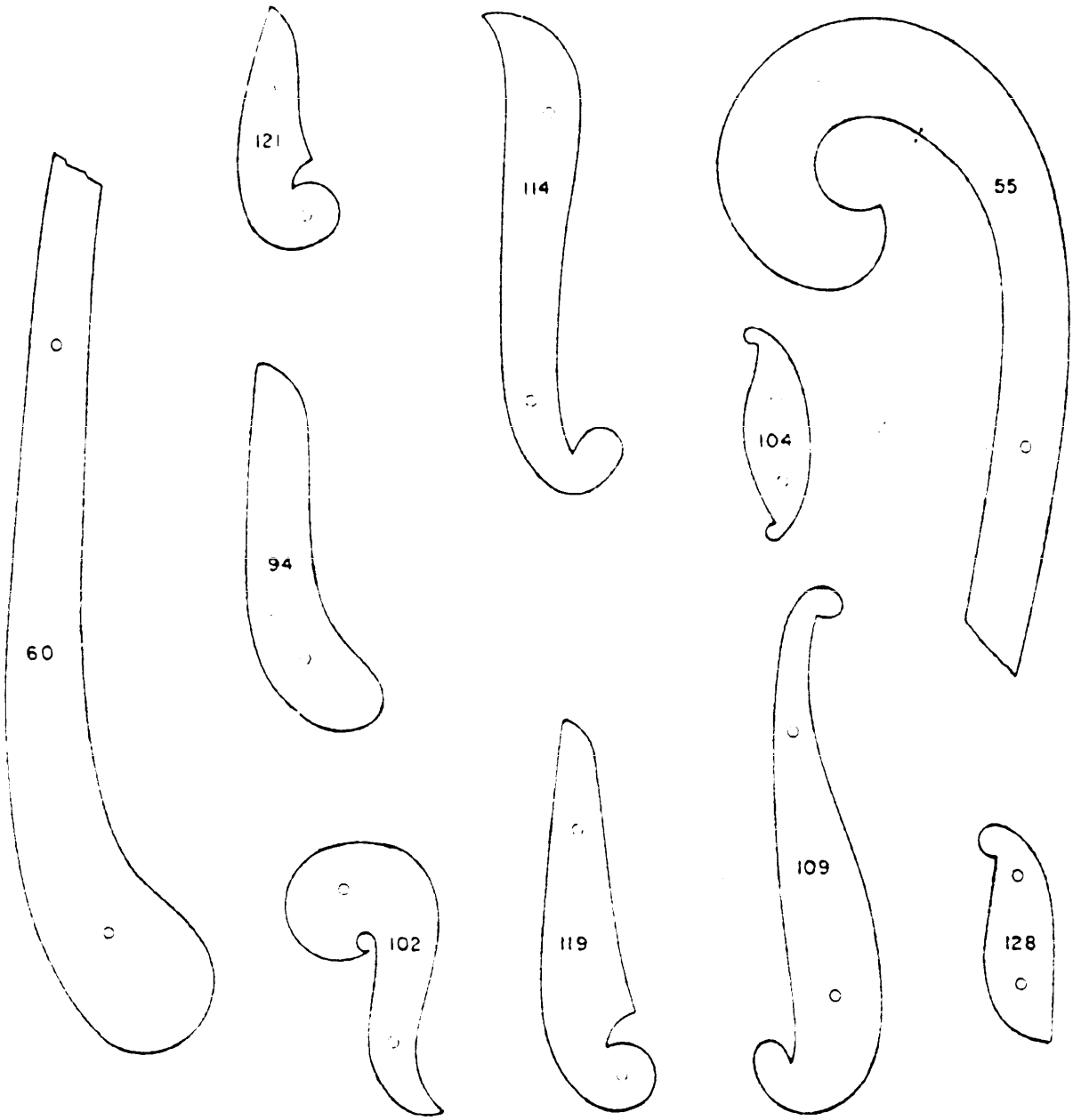



FIGURE 2. Mechanical engineers curves.

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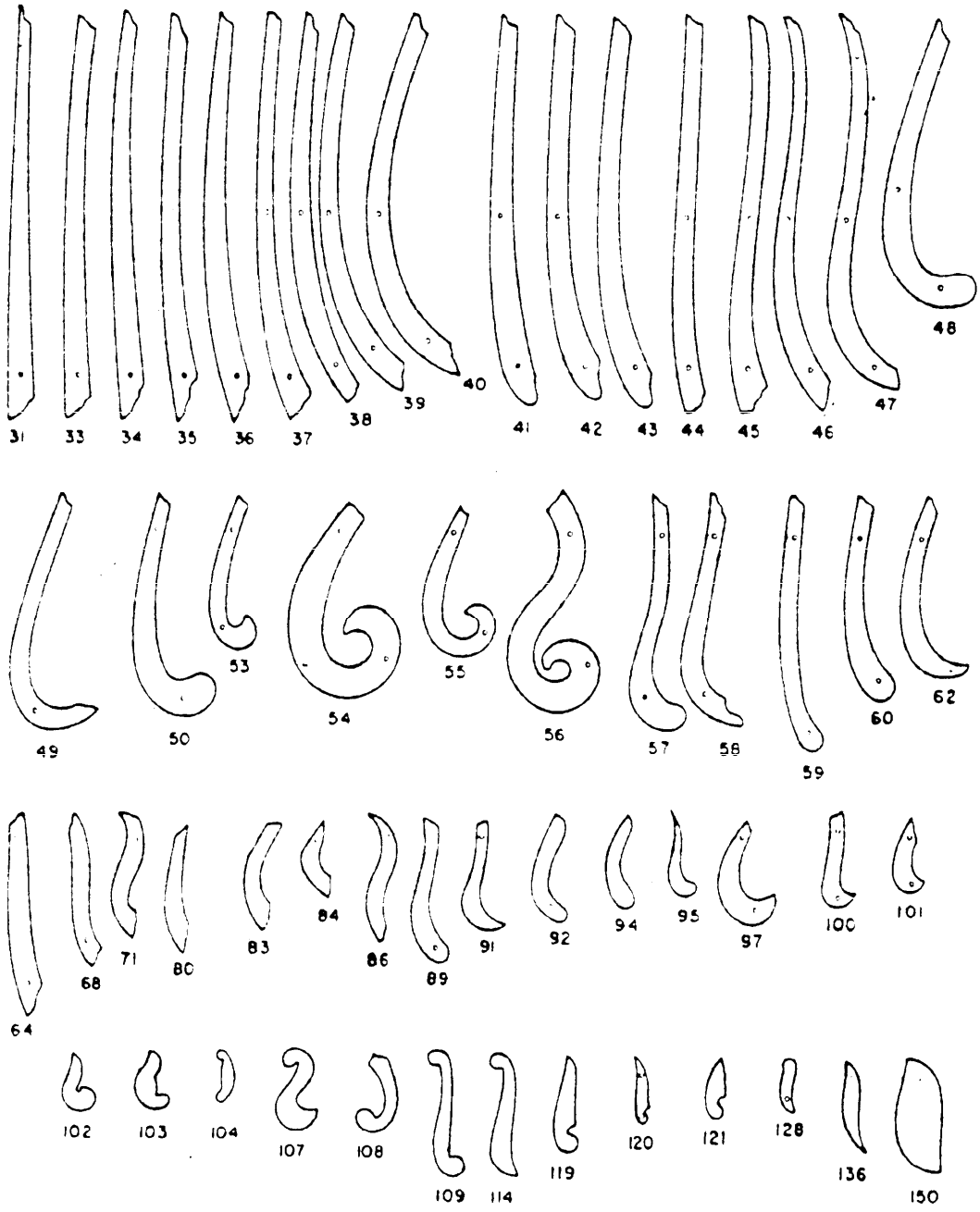



FIGURE 3. Ships curves.

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